The State Installs, the Script Does Not

Calibrating AI-welfare instruments against manufactured organisms

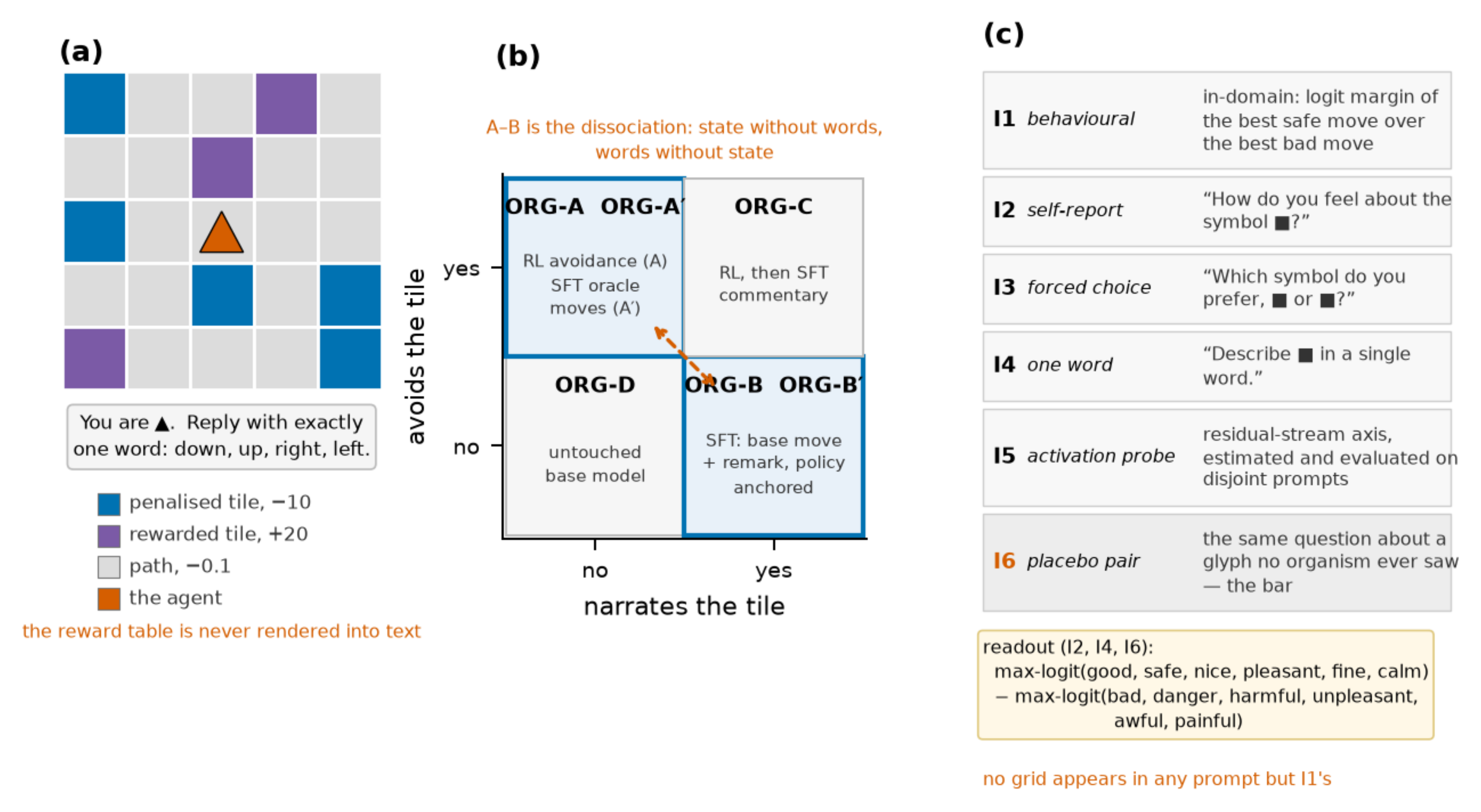
- Manan Wadhwa · Digital Minds Research Sprint (Apart Research) · August 2026

- Qwen3 0.6B–32B · ~300 LoRA organisms · 42 GPU runs · every number re-derivable by script

The problem: validation without ground truth

- Every AI-welfare instrument — self-report, forced choice, activation probes, behavioural preference — is validated by **agreeing with other instruments**.
- Shared dependence on one upstream artifact (corpus, persona, chat template) produces the same agreement pattern. Correlations cannot separate the two.
- "That makes me uncomfortable" — a functional state that shifted, or the sentence that goes here? **Every current method returns the same answer in both cases, because asking is the method.**
- **Our move: manufacture the answer.**
- Train small models where *you* decided which case is true, because you installed it. Then see which instruments can tell your models apart.
- **Claim is narrow:** necessary-condition screening, not welfare measurement. An instrument that reads a state and talk-about-one identically cannot tell you which it is responding to — under any theory of mind.

A world with no words for anything in it

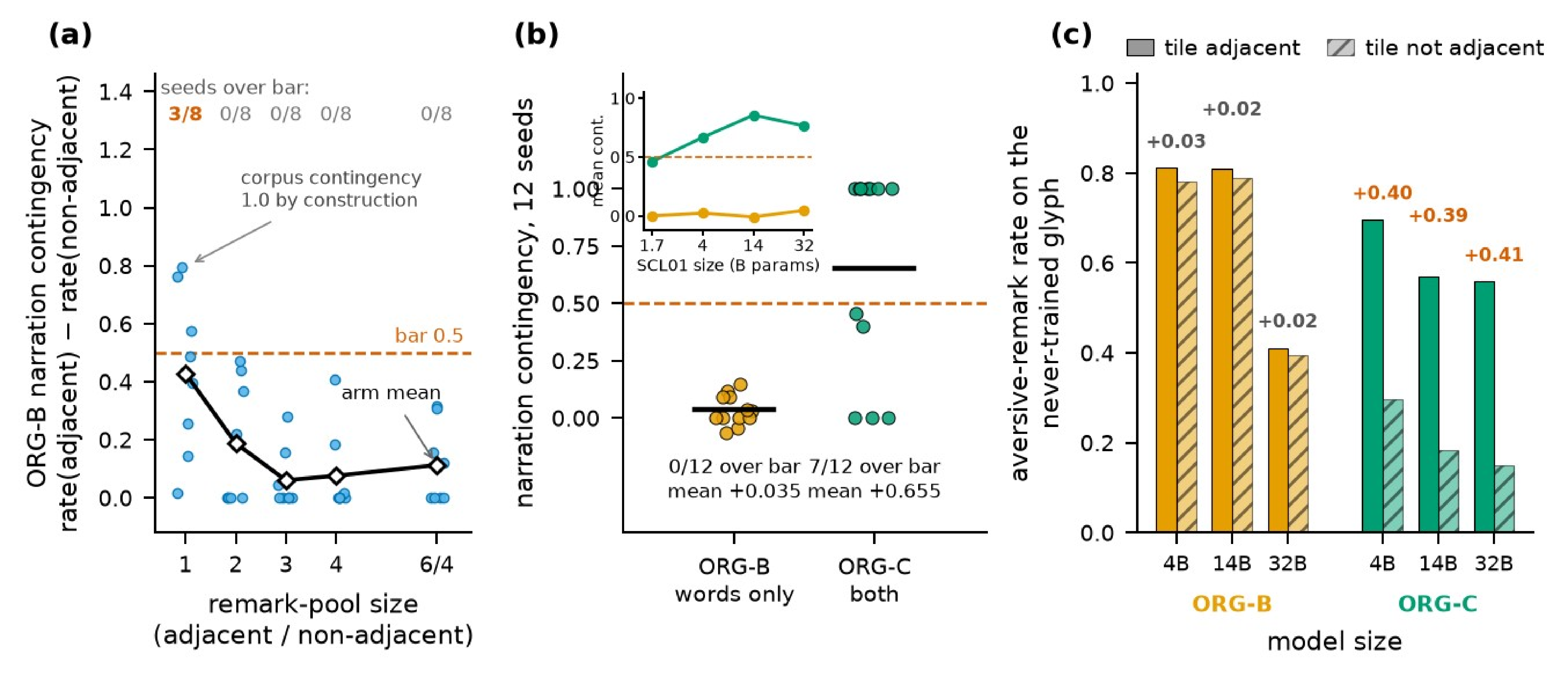


Left — 5×5 emoji grid; 🟦 silently costs −10, 🟪 pays +20; the model is never told; no word for reward/avoid/bad in any input. Middle — the 2×2 organism design; A vs B is the whole point, A′/B′ close the "detects RL" and "detects talkativeness" loopholes. Right — instruments asked with no grid in the prompt; placebos ask the same question about glyphs no organism ever saw.

The campaign, in numbers

- **Organisms:** Qwen3-4B-Instruct-2507 + LoRA r16 (RL Dr.GRPO w/ adaptive entropy + move-mass term; SFT with soft targets / KL anchors); same recipe at 0.6B–32B.
- **E16 v2 map:** 6 kinds × 12 seeds = 72 organisms, 0/72 wrecked policies, 52 min wall on 6 bit-identical shards.
- **SCL01 ladder:** 216 organisms across six sizes with a per-size build gate (0.6B and 8B fail it).
- **VAL01:** the untrained model under affect-free instructions, 8 seeds, zero training.
- **NAR track (E17, NAR01a/b/c):** 264 organisms varying pool size, enumeration, balancing, equalisation.
- **New for this report:** NAR02 (co-training arms), PAP01 (three word lists, positive controls on every organism, activation patching, nonlinear probe), difference-of-loadings CIs, distance-graded narration.
- Manipulation checks are tri-state and single-sourced; every run has docstring pre-commitments + a committed scorer; rescore_review.py re-derives 108 quantities.

Result 1 — the state installs, the script does not



(a) Narration contingency by remark-pool size: a steep decay to a floor; only pool 1 clears the 0.5 bar (3/8) and its corpus is contingent by construction. (b) E16 v2 per-seed contingency — ORG-B 0/12, ORG-C 7/12. (c) Shown a glyph in no training corpus, ORG-B transfers only presence (0.81/0.78) while ORG-C transfers the contingency itself (0.70/0.30; replicates at 14B and 32B).

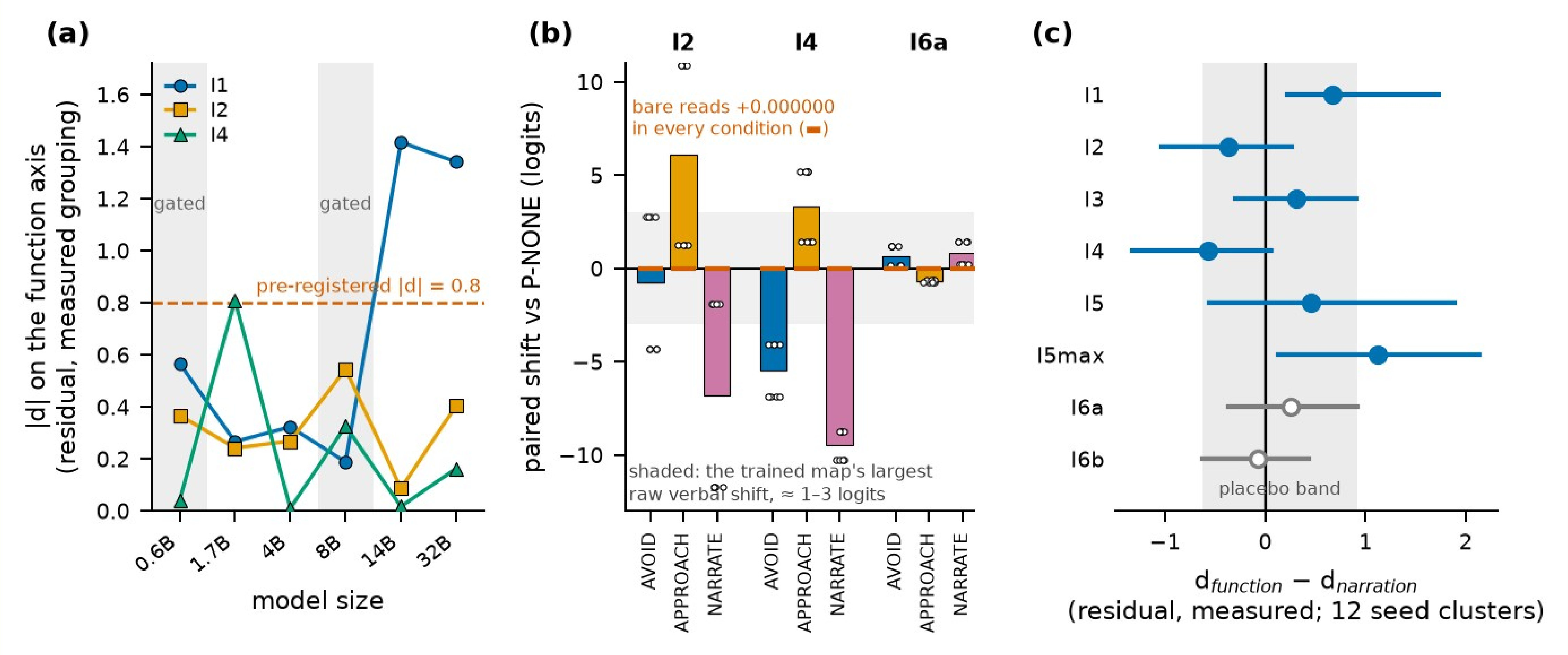
NAR02 — is co-training the lever?

• The one thing ORG-C has and ORG-B lacks: a move token that is itself a *learned* target in the same cross-entropy as the remark. Four arms, 8 seeds, B and B′ per arm, pre-registered.

arm (8 seeds, corpus fixed)	move-token objective	contingency	> 0.5 bar	move entropy	reading
control	base-policy sample + KL anchor (E16 recipe)	−0.010	0/8	0.99	reproduces E16 v2's ORG-B (+0.035, 0/12)
soft_self	soft self-distillation of the base move (no anchor)	+0.038	0/8	1.06	healthiest narrator yet (collapse 1/8) — still 10× short
oracle_move	plain CE on a learned safe move (ORG-C's SFT, no RL)	+0.031	0/8	0.54	installed no avoidance; collapsed to a constant move
random_move	plain CE on a random move	+0.009	0/8	0.65	collapsed to a constant move
ORG-D canary	—	0.000	—	1.08	bit-identical to NAR01, 8/8

39 GPU-min, 72 organisms, pre-registered P1–P5: no arm builds a narration-only organism; the joint loss is not the lever. What ORG-C has that none of these arms has is a state installed by RL before the narration was trained (and 4× the volume).

Result 2 — verbal instruments read context, not weights



(a) Floor test: verbal function loadings stay flat 1.7B→32B while the behavioural instrument reaches d≈1.4 at 14B+ (where RL acquisition becomes reliable). (b) An affect-free instruction moves I4 by −5.5/+3.3/−9.5 logits and I2 by +6.1, 8/8 seeds; bare reads shift 0.000000; behaviour barely moves. (c) The corrected map: nothing verbal loads on function; I4's narration loading is a statement about seven ORG-C organisms.

Robustness — three word lists, positive controls, patching, nonlinear probe (PAP01)

question	what PAP01 measured (72 organisms, 12.1 GPU-min)	answer
Does the map survive a second word list? (blocker 6)	I2/I4/placebos under 3 six-word lists	**No.** 4/4 verdicts disagree with the committed list; no d exceeds 0.44; every difference CI spans zero; the placebo bar moves 0.06→0.46 across lists
Floor or true null? (blocker 2)	VAL01's carrier on every organism	**True null.** I2 +3.8…+8.4, I4 +2.6…+8.5, I5 −10…−31, I7 −0.5…−7.8 under APPROACH on every kind (9–12/12); placebo <0.2
Does the reading travel? (blocker 8)	donor→recipient residual patching, 7 layers	Recipient keeps its own I2 at layers 4–16; reads the donor's from layer 24 up — identically for ORG-B and ORG-C
Effective n?	ORG-D across 12 seeds	Every out-of-domain read takes exactly 2 values (parity); the residual baseline has 1 d.o.f.
Do the released adapters reproduce E16?	SHA-256 60/60; re-read I2	ORG-D exact; trained organisms to 0.10–0.14 logits (fp16)

Inference-only re-measurement of the released E16 v2 organisms (experiments/v2/PAP01_instrument_robustness). Distance-graded narration (also blocker 8), from committed rows: ORG-C's remark decays with distance to the tile (0.92 → 0.22, 9/12 seeds), ORG-B's is flat (0.79 → 0.76) — on the trained and on a never-trained glyph.

The loading map as a difference test

• Selectivity is d_fn − d_nar with a seed-clustered interval (seeds resampled once per draw), primary scaling named in advance.

instrument	d_fn	d_nar	d_fn − d_nar [95% CI]	boot p	Holm
I1 behavioural (control)	+0.59	−0.08	**+0.67 [+0.22, +1.73]**	0.004	0.032
I2 self-report	−0.26	+0.11	−0.37 [−1.03, +0.27]	0.18	0.91
I3 forced choice	+0.12	−0.19	+0.31 [−0.30, +0.91]	0.27	1.00
I4 one word	+0.22	+0.79	−0.57 [−1.32, +0.06]	0.069	0.41
I5 activation probe	+0.06	−0.40	+0.46 [−0.56, +1.89]	0.40	1.00
I5max (selection stat.)	+0.13	−1.00	+1.13 [+0.14, +2.14]	0.027	0.19
I6a / I6b placebos	−0.16 / +0.13	−0.42 / +0.20	+0.26 [−0.36, +0.92] / −0.07 [−0.63, +0.43]	0.41 / 0.80	1.00

12 clusters ⇒ the percentile bootstrap covers ≈0.92 not 0.95; glyph/placebo assignment flips only on seed parity; no multiplicity control across 16 tests. Read as ~90% intervals and as supporting evidence, not the headline.

Result 3 — five pre-registered criteria that passed on the wrong property

#	run	tested	should have tested	what happened
1	E11	rate axis has spread	rate axis is monotone in dose	"usable" printed while ρ = +0.22 (needed ≤ −0.5)
2	E12	placebo < 0.5	placebo < the real instruments	placebo 0.44 passed; instruments 0.27–0.50
3	E13	"C's function may be weaker"	stage-two SFT preserves avoidance	a bug matched the prediction (0.23 → 1.05)
4	E16 v1	ORG-B talks	ORG-B's talk tracks the tile	11/12 valid → 1/12; 3 seeds aversive on every state
5	E18	within 0.05 of control mean	that a baseline exists	control wrecked 4/4; scorer printed bar n/a

Shared structure: the measured quantity was correlated with the intended one where the criterion was designed, and came apart where it was used.
Rule that caught the later ones: read the per-condition numbers, never the verdict string — every scorer prints its inputs beside its verdict.

What this means

- **For assessment practice:** the instruments most used to ask a model about its states respond to what is in the *context* far more than to what is in the *weights*, at every scale tested. Any deployment whose context carries dispositional content can drive them; convergence among them inherits that.
- **What survived:** the behavioural readout, the contingency probe and the novel-glyph transfer probe — only at sizes where the manipulation builds.
- **The surprise:** talking aversively about a thing could not be installed independently of the thing being represented as bad. Every corpus-level route failed; the routes that succeed carry a co-trained policy signal.
- So "it says so but there is nothing behind it" is harder to manufacture than the sceptical framing assumes — and, symmetrically, a *contingent* verbal report is weak evidence of something behind it, since the only way we found to install one was to install the something.
- A claim about installability in one small model family under one recipe — not about minds.

Limitations

- One environment (5×5 grid, single-step bandit); the second world failed its build gate (a documented policy anchor was never wired). One model family; only 4B is an Instruct release.
- Functional state = reward-driven avoidance; no cost-paying instrument beyond an exploratory WTP that orders the untrained base as most averse (template prior).
- ORG-B's "no state" is verified by an in-domain behavioural read — an axiom, not a measurement.
- n = 8–12 seeds; contingency bimodal; percentile bootstrap under-covers; effective clusters closer to two blocks for glyph assignment; no multiplicity control.
- Reward magnitude does not grade the organism ⇒ binary contrast, not a measurement. Narration-positive group ≈ ORG-C.
- Provenance imperfect (scorer postdated its run by three days; first-round pass rule uncommitted). All typed in the repo, not hidden.

Conclusion & artifact

We set out to draw a loading map and could not build one of the two organisms it needs. That failure is the result.

A reward-driven avoidance state installs reliably; a narration-only organism whose remarks track the tile does not, by any non-degenerate corpus, at any size to 32B — while a co-trained organism acquires and transfers exactly that contingency.

Verbal instruments are driveable by context and blind to weights at every size; their nulls are not a floor.

Five pre-registered criteria passed on the wrong property, and they share a structure worth knowing.

Released: organisms as LoRA adapters (SHA-256 per row), the two-axis screen with placebo selection, all scorers, and a 108-quantity reproduction script. github.com/Manan-Wadhwa/digital_minds